

Recursive Time-Crystalline Memory on a Marked Causal Network

A minimal falsifiable toy model linking recursive phase memory, ancestry transfer, and robust subharmonic order

Abstract

This note tests a narrow claim: whether the recursive-memory ideas previously developed for AI.Web can be translated into a minimal dynamical model that meets nontrivial time-crystal-like criteria rather than merely repeating periodically. The model treats each recursive generation as a new causal layer, each node as carrying a binary internal phase mark, the ancestry-transfer step as an imperfect global phase inversion, and locality weights w_{ij} as coupling marks across adjacent generations. Numerical experiments on a 300-node sparse random network show a clear transition from disordered dynamics to robust period-doubled order as coupling increases. At 12% ancestry-transfer error, the staggered long-time order parameter rises from approximately 0.057 at $K=1.4$ to 0.325 at $K=1.6$ and 0.670 at $K=1.8$, reaching 0.893 at $K=2.4$. At $K=2.4$, the ordered response survives transfer-error rates through about 15%, weakens strongly near 20%, and collapses by 25%. This is not evidence that spacetime is a time crystal, nor that the 1–9 FBSC architecture is a fundamental law of physics. It is evidence that the core idea of a phase-locked recursive memory lattice can be expressed as a conventional dynamical-systems model exhibiting robust subharmonic order and a disorder-to-order transition.

1. Motivation

Earlier AI.Web work described recursive identity as a loop that returns without requiring microscopic identity: $\psi_i' \neq \psi_i$ while an invariant identity can persist. It also introduced phase-locked memory lattices, resonance-gated edges, recursive re-entry, persistence thresholds, and an ancestry-transfer operator. The key question is whether those concepts can be stripped of symbolic language and written as a model with ordinary state variables, interactions, order parameters, perturbations, and falsifiable failure conditions.

Modern time-crystal physics provides a useful comparison because a discrete time crystal is not simply a periodic oscillator. Its defining feature is robust temporal order: a system driven with period T can respond at an integer multiple qT , with that subharmonic response remaining rigid over a range of perturbations. Recent work has also connected time-crystalline dynamics with internal clocks, delayed feedback, space-time order, and computational memory.

2. Minimal marked causal-network model

Let generation n contain N nodes. A node is labeled $x=(i,n)$, where i identifies the lineage/site and n is the recursion step. Causal adjacency is restricted to neighboring generations, so edges point from generation n to generation $n+1$. This makes the unfolded history acyclic even though the internal phase pattern can recur.

Each node carries an internal binary phase mark
 $s_i(n) \in \{-1,+1\}$.

The geometric/event layer and the internal mark layer are intentionally separated. This preserves the possibility of a Poisson-like causal substrate while allowing temporal order to develop in the marks rather than in the event locations themselves.

The locality structure is represented by weights
 $w_{ij} \geq 0$,
with $w_{ij}=0$ for unrelated lineages and normalized local field
 $h_i(n) = \sum_j w_{ij} s_j(n) / \sum_j w_{ij}$.

3. Ancestry-transfer operator

One recursive step has two stages.

Stage A: imperfect phase transfer.

The intended ancestry operation is a phase inversion,

$$s_i \rightarrow -s_i,$$

but each site fails to invert with probability ϵ .

This is the toy analogue of an imperfect recursive handoff. Exact inversion would trivially create period doubling, so robustness must be tested at $\epsilon > 0$.

Stage B: resonance/locality correction.

After the imperfect transfer, the next-generation mark is selected from the local ancestral field using

$$P[s_i(n+1)=+1] = 1 / (1 + \exp[-2K h_i(n)]).$$

K is the effective resonance/coupling strength. Small K allows local errors to proliferate. Large K causes neighboring ancestry channels to mutually correct and phase-lock.

In compact form, the stochastic transfer operator is

$$S(n+1) = \mathcal{A}_{K,\epsilon}[S(n)].$$

The full unfolded graph remains causal across generations; recurrence occurs in internal state space, not through backward causal edges.

4. Time-crystalline order parameter

Ordinary recurrence is insufficient. We therefore define the period-doubled staggered order parameter

$$Q_2 = \left| \lim_{M \rightarrow \infty} \left(\frac{1}{M} \sum_n (-1)^n m(n) \right) \right|,$$

where

$$m(n) = \frac{1}{N} \sum_i s_i(n).$$

If the system is disordered, Q_2 approaches zero. If the recursive lattice develops stable period-2 order, Q_2 remains nonzero.

A second diagnostic is the fraction of post-burn-in spectral power concentrated at the Nyquist/subharmonic frequency π . A robust ordered phase should show both nonzero Q_2 and a sharp spectral line rather than merely transient oscillation.

5. Numerical experiment

The initial experiment used:

$N = 300$ nodes

mean initial neighborhood size ≈ 8 before symmetrization

400 recursive generations

150-generation burn-in

12 independent runs per coupling value for the coupling sweep

ancestry-transfer error $\varepsilon = 0.12$

The network was sparse and randomly connected. The same locality graph was used through the recursive layers, which is equivalent to unfolding a fixed weighted neighborhood rule across generations.

6. Result: disorder-to-recursive-time-crystal transition

Observed mean staggered order Q_2 :

$K=0.2$	$Q_2=0.0038$
$K=0.8$	$Q_2=0.0095$
$K=1.2$	$Q_2=0.0193$
$K=1.4$	$Q_2=0.0572$
$K=1.6$	$Q_2=0.3247$
$K=1.8$	$Q_2=0.6703$
$K=2.0$	$Q_2=0.7845$
$K=2.4$	$Q_2=0.8931$
$K=3.0$	$Q_2=0.9541$
$K=4.0$	$Q_2=0.9849$

The transition region lies roughly between $K \approx 1.4$ and $K \approx 1.8$ for this finite system and protocol. The spectral diagnostic changes at the same place: mean subharmonic power fraction rises from about 0.20 at $K=1.4$ to 0.70 at $K=1.6$ and 0.99 at $K=1.8$.

This is important because the system was not given perfect recurrence. At $\varepsilon=0.12$, every generation contains a substantial transfer defect. The collective coupling repairs enough of those defects that the global trajectory locks into a persistent 2T pattern.

7. Robustness to ancestry-transfer error

Holding $K=2.4$ fixed:

$\varepsilon=0.00$	$Q_2=0.9817$
$\varepsilon=0.05$	$Q_2=0.9621$
$\varepsilon=0.10$	$Q_2=0.9216$
$\varepsilon=0.15$	$Q_2=0.8313$
$\varepsilon=0.20$	$Q_2=0.4347$
$\varepsilon=0.25$	$Q_2=0.0506$
$\varepsilon=0.30$	$Q_2=0.0253$
$\varepsilon=0.40$	$Q_2=0.0068$

Thus the phase is not merely the exact-inversion solution. It occupies a finite perturbative region. In this toy system the recursive temporal order degrades strongly around $\epsilon \approx 0.20$ and is effectively lost by $\epsilon \approx 0.25$.

8. Mapping to the earlier recursive framework

The correspondence is:

recursive generation $\rightarrow$ discrete time/Floquet step
ancestry-transfer operator $\rightarrow$ imperfect phase inversion plus inherited local field
phase-locked memory lattice $\rightarrow$ synchronized attractor across internal phase marks
resonance strength $\rightarrow$ coupling K
drift/error $\rightarrow$ transfer defect ϵ
persistence $\rightarrow$ nonzero long-time Q_2 and narrow subharmonic spectral response
identity through change $\rightarrow$ invariant trajectory class rather than identical microscopic state
 w_{xy} locality $\rightarrow$ weighted cross-generation influence
sealed recursive loop $\rightarrow$ stable orbit of the transfer operator
octave return $\rightarrow$ return of an invariant after q recursive steps

This suggests a cleaner formal statement of recursive identity:

memory is not necessarily a stored configuration $S(n_0)$;
memory can be a stable equivalence class of trajectories $[S(n)]_A$ under the ancestry-transfer dynamics.

9. Relation to current time-crystal physics

The toy model is structurally consistent with several established directions in time-crystal research, while remaining distinct from them.

Time-crystal clocks: recent theory shows that time-crystalline dynamics can function as genuine quantum clocks, supporting the narrower idea that persistent internal recurrence can provide an internal temporal reference.

Retarded feedback: spin-maser experiments have demonstrated a time-crystal phase produced by delayed feedback, including an emergent self-sustained frequency and a threshold in

feedback strength. This gives a direct physical precedent for delay/feedback creating robust temporal order.

Space-time crystals: experiments have demonstrated systems that jointly break spatial and temporal translation symmetries. These are ordered phases inside spacetime; they do not imply that spacetime itself is a time crystal.

Computational memory: discrete-time-crystal dynamics have been used experimentally as a quantum reservoir, with measurable memory, nonlinear, and information-scrambling capacities. This is especially relevant to the AI.Web hypothesis that persistent phase dynamics can serve as a memory/computation substrate.

10. What this model does not show

This toy model does not establish that:

spacetime emerges from time crystallization;

the universe is a time crystal;

the 1–9 FBSC sequence is fundamental;

Tesla-style 3–6–9 harmonics are privileged by known time-crystal physics;

the previously derived $d_{\text{eff}}=4$ mechanism follows from this model;

the system is a thermodynamic phase of matter in the rigorous many-body sense;

or the present classical simulation constitutes experimental evidence.

The current result is narrower: a recursive ancestry network with imperfect transfer and locality-based correction can develop a robust, subharmonic dynamical attractor separated from a disordered regime by a coupling-controlled transition.

11. Why separating geometry from phase marks matters

The spacetime program previously used Poisson-like event statistics and higher-order factorial cumulants. Direct periodic ordering of event locations could conflict with those assumptions by introducing long-range temporal correlations into the event process itself.

The marked-network construction avoids that problem. Let the event substrate remain statistically compatible with the causal/Poisson construction, while each event carries an

internal phase mark θ_x or s_x . Temporal crystallization then occurs in the correlation structure of the marks.

A future generalized model can therefore take the form:

causal event network + internal phase marks + w_{xy} locality + ancestry transfer $\rightarrow$ ordered recursive temporal field.

That is a stronger and cleaner hypothesis than spacetime itself = time crystal.

12. Immediate next tests

The present positive result is only a first filter. The model should now be challenged by:

- finite-size scaling to determine whether the transition sharpens or disappears with N ;
- randomized and time-varying w_{ij} rather than a frozen graph;
- strictly directed locality weights derived from the causal order;
- $q=3$ through $q=9$ transfer operators to test whether stable higher-order recursive periods exist;
- comparison against a finite-state recurrence null model;
- autocorrelation lifetime and susceptibility near the apparent transition;
- information-retention tests that encode a perturbation or symbolic bit pattern and measure recoverability;
- removal of the imposed inversion drive to distinguish driven discrete order from autonomous oscillation;
- and integration with the existing W_N / factorial-cumulant spacetime machinery without violating its Poisson fixed-point assumptions.

13. Working hypothesis

A useful working hypothesis is:

A recursive memory lattice can be modeled as a marked causal network whose internal phase variables undergo imperfect ancestry transfer and local resonance correction. Above a critical coupling, collective correction may stabilize a subharmonic trajectory class that persists despite transfer errors. Such a trajectory class provides a concrete dynamical interpretation of phase-locked memory and recursive identity.

This hypothesis is falsifiable. It can fail under finite-size scaling, altered topology, causal-direction constraints, noise, removal of the drive, or information-capacity tests.

14. Provisional conclusion

The minimal experiment survives the first falsification attempt. The result does not validate the cosmological interpretation of recursive time crystals, but it does validate a narrower mathematical engineering concept: phase-locked recursive memory can be represented by a conventional stochastic dynamical system that exhibits robust period-doubled order, a coupling-dependent transition, and finite tolerance to ancestry-transfer defects.

That is sufficient to justify further study and to preserve the model as a formal research direction rather than leaving it as chat-only speculation.

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